

EXPERIMENT NO 12

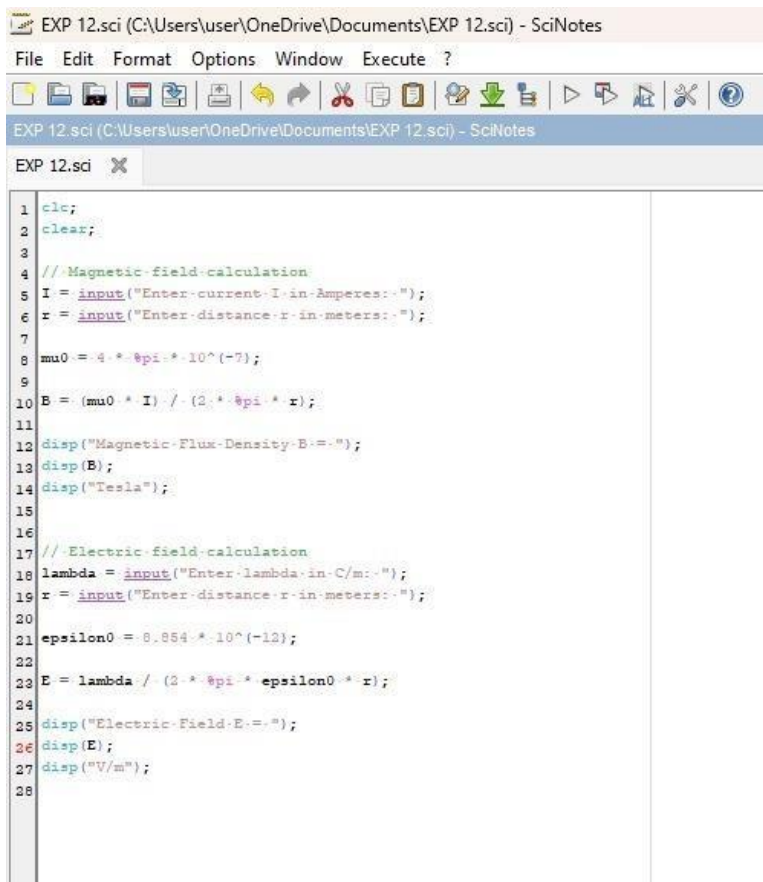
Determination and plotting of magnetic and electric field intensity due to infinite long conductor at any specific point located at H

Aim:

To write program for Magnetic field intensity of infinite long conductor using SCILAB.

Software required:

- SCILAB version 6.0.1 **Program:**



```
EXP 12.sci (C:\Users\user\OneDrive\Documents\EXP 12.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 12.sci (C:\Users\user\OneDrive\Documents\EXP 12.sci) - SciNotes
EXP 12.sci
1 clc;
2 clear;
3
4 // Magnetic field calculation
5 I = input("Enter current I in Amperes:");
6 r = input("Enter distance r in meters:");
7
8 mu0 = 4 * %pi * 10^(-7);
9
10 B = (mu0 * I) / (2 * %pi * r);
11
12 disp("Magnetic Flux Density B = ");
13 disp(B);
14 disp("Tesla");
15
16 // Electric field calculation
17 lambda = input("Enter lambda in C/m:");
18 r = input("Enter distance r in meters:");
19
20 epsilon0 = 8.854 * 10^(-12);
21
22 E = lambda / (2 * %pi * epsilon0 * r);
23
24 disp("Electric Field E = ");
25 disp(E);
26 disp("V/m");
27
28
```

Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\user\Documents\

Name

- Documents
- ..
- My Music
- My Pictures
- My Videos
- ex 6.scd
- EXP 12.scd~

File/directory filter

Scilab 2026.1.0 Console

Enter current I in Amperes: 5

Enter distance r in meters: 3

"Magnetic Flux Density B = "

0.0000003

"Tesla"

Enter lambda in C/m: 4

Enter distance r in meters: 5

"Electric Field E = "

1.438D+10

"V/m"

-->

Variable Browser

Name	Value	Type	Visibility	Memory
B	3.33e-07	Double	local	216 B
E	1.44e+10	Double	local	216 B
I	5	Double	local	216 B
epsilon0	8.85e-12	Double	local	216 B
lambda	4	Double	local	216 B
mu0	1.26e-06	Double	local	216 B
r	5	Double	local	216 B

Command History

```
// -- 10/08/2026 14:50:34 -- //
// -- 24/08/2026 04:27:36 -- //
// -- 24/08/2026 04:29:27 -- //
-3*10^-7
6
-4
-2
5
3
-4
-5
-4
-5
```

News feed

Discover Scilab community

Scilab is open source.

Giving back support or use cases is appreciated.

- Add and improve functions on [Gistlab](#).
- View and edit [help pages](#).
- Publish toolboxes on [ATCORS](#).
- Post questions on [Discourse](#).

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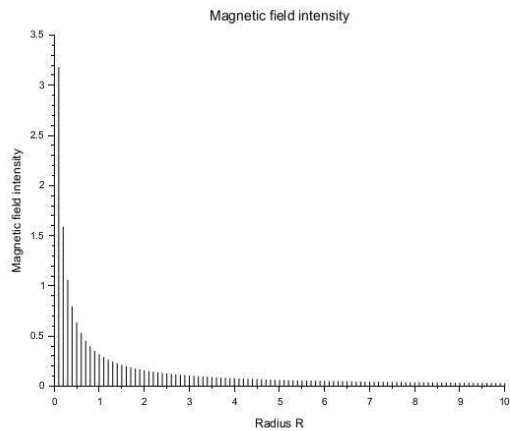
ENG IN

4:59 AM 8/24/2026

Graphic window number 1

File Tools Edit ?

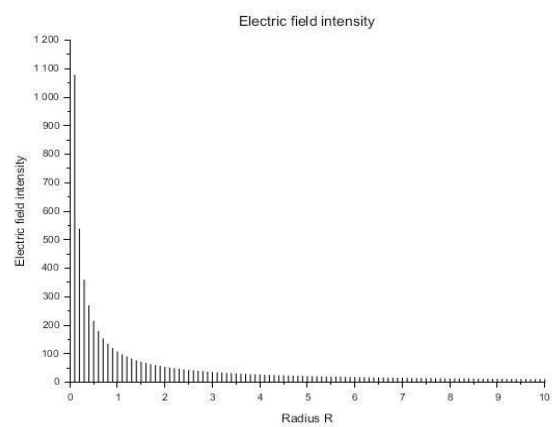
Graphic window number 1



Graphic window number 2

File Tools Edit ?

Graphic window number 2



Theoretical calculation:

Ex no. 12

$$current = 3A$$

$$radius = 3\text{ mm} = 0.003\text{ m}$$

$$x = 2\text{ m}$$

$$radius = 5\text{ mm}$$

1. Magnetic field intensity (H)

$$H = \frac{i}{2 \times 3.14 \times r} = \frac{3}{2 \times 3.14 \times 0.003} = \frac{3}{0.01884} = 159.23\text{ A/m}$$

2) Electric field intensity (E)

$$E = \frac{h}{2 \times 3.14 \times \epsilon_0 \times r}$$

$$\epsilon_0 = 8.85 \times 10^{-12}, \quad r = 5\text{ mm}$$

$$x = 2$$

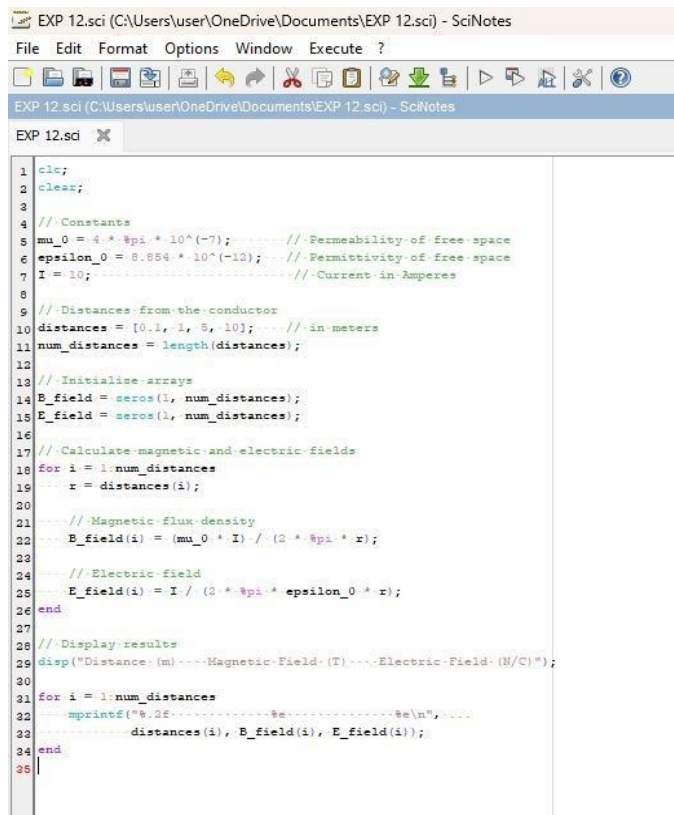
$$Denominator = 2 \times 3.14 \times 8.85 \times 10^{-12} \times 0.005$$

$$= 2.780156 \times 10^{-13}$$

$$E = \frac{2}{2.780156 \times 10^{-13}} = 7.1938 \times 10^{12}\text{ V/m}$$

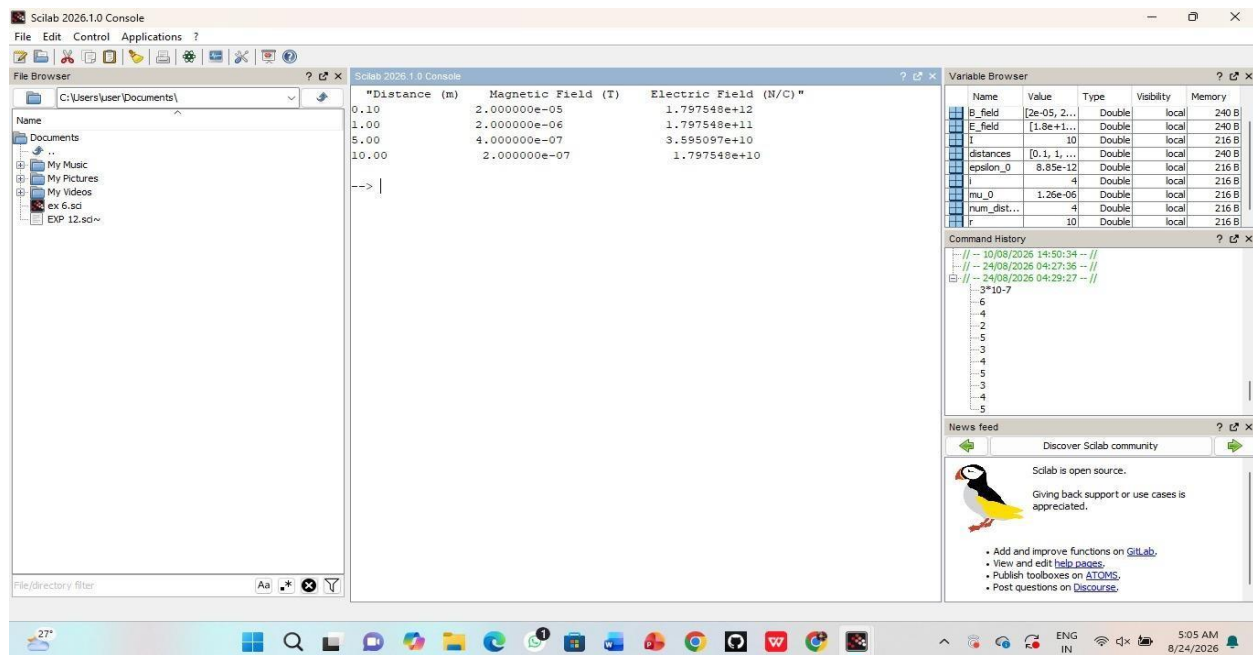
Case 1: To examine how the magnetic and electric field intensities change with varying distances from an infinitely long current-carrying conductor.

Program:



```
1  clc;
2  clear;
3
4  // Constants
5  mu_0 = 4 * pi * 10^(-7); // Permeability of free space
6  epsilon_0 = 8.854 * 10^(-12); // Permittivity of free space
7  I = 10; // Current in Amperes
8
9  // Distances from the conductor
10 distances = [0.1, 1, 5, 10]; // in meters
11 num_distances = length(distances);
12
13 // Initialize arrays
14 B_field = zeros(1, num_distances);
15 E_field = zeros(1, num_distances);
16
17 // Calculate magnetic and electric fields
18 for i = 1:num_distances
19     r = distances(i);
20
21     // Magnetic flux density
22     B_field(i) = (mu_0 * I) / (2 * pi * r);
23
24     // Electric field
25     E_field(i) = I / (2 * pi * epsilon_0 * r);
26 end
27
28 // Display results
29 disp("Distance (m) --- Magnetic Field (T) --- Electric Field (N/C)");
30
31 for i = 1:num_distances
32     fprintf("%2f --- %2f --- %2f\n", distances(i), B_field(i), E_field(i));
33 end
34
35
```

Output:

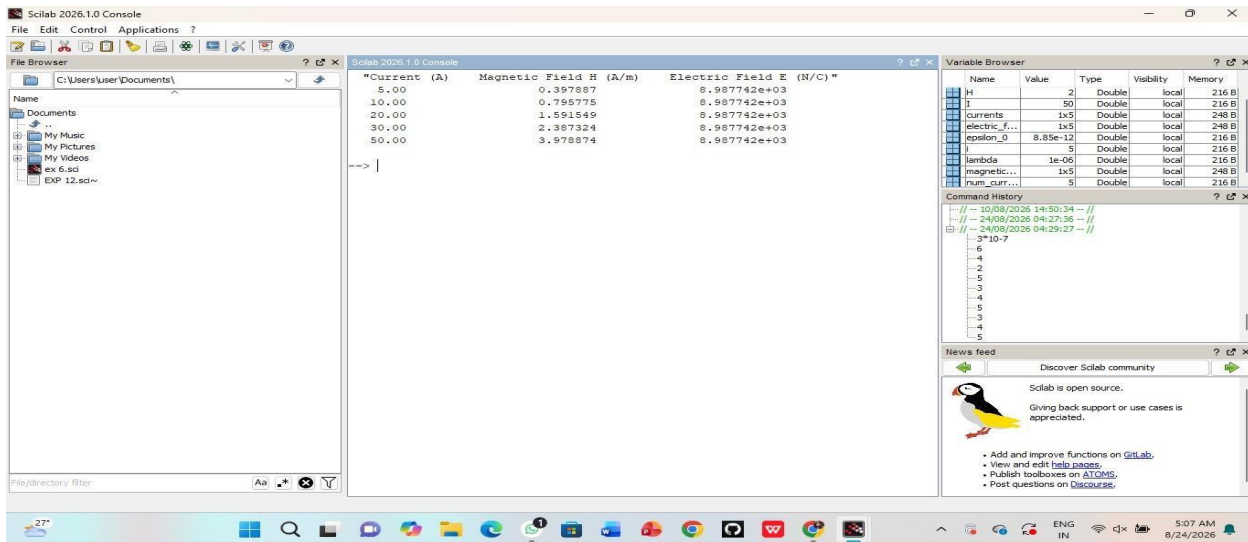


Case 2: To study how the magnetic and electric field intensities change for an infinitely long conductor at a fixed distance H but with varying currents.

Program:

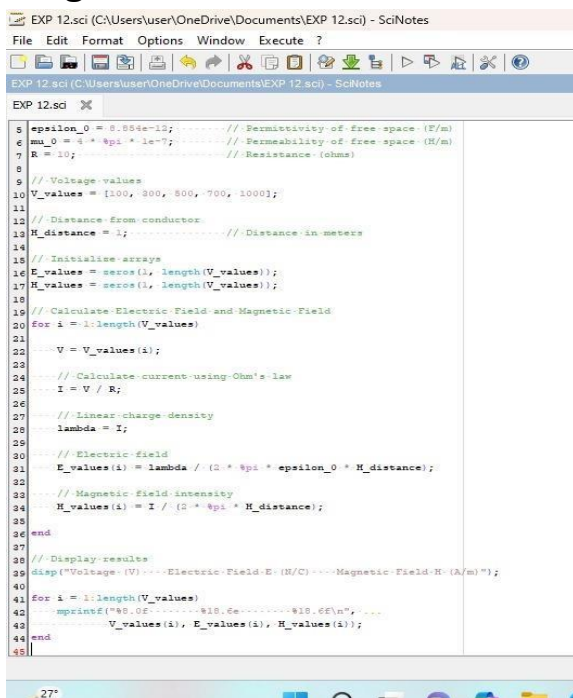
```
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File Edit Format Options Window Execute ?
EXP 12.sci (C:\Users\User\OneDrive\Documents\EXP 12.sci) - SciNotes
EXP 12.sci
1 clear;
2 clear;
3
4 // Constants
5 H = 2; // Distance from conductor (m)
6 epsilon_0 = 8.854e-12; // Permittivity of free space (F/m)
7 lambda = 1.0e-6; // Linear charge density (C/m)
8
9 // Currents to evaluate
10 currents = [5, 10, 20, 30, 50];
11 num_currents = length(currents);
12
13 // Preallocate arrays
14 magnetic_field_intensity = zeros(1, num_currents);
15 electric_field_intensity = zeros(1, num_currents);
16
17 // Calculate magnetic and electric field intensities
18 for i = 1:num_currents
19     I = currents(i);
20
21     // Magnetic field intensity
22     magnetic_field_intensity(i) = I / (2 * %pi * H);
23
24     // Electric field intensity
25     electric_field_intensity(i) = lambda / (2 * %pi * epsilon_0 * H);
26
27 end
28
29 // Display results
30 disp('Current (A) --- Magnetic Field H (A/m) --- Electric Field E (N/C)');
31
32 for i = 1:num_currents
33     mprintf('%8.2f --- %8.2f --- %8.2f\n', ...
34             currents(i), ...
35             magnetic_field_intensity(i), ...
36             electric_field_intensity(i));
37 end
38
```

Output:

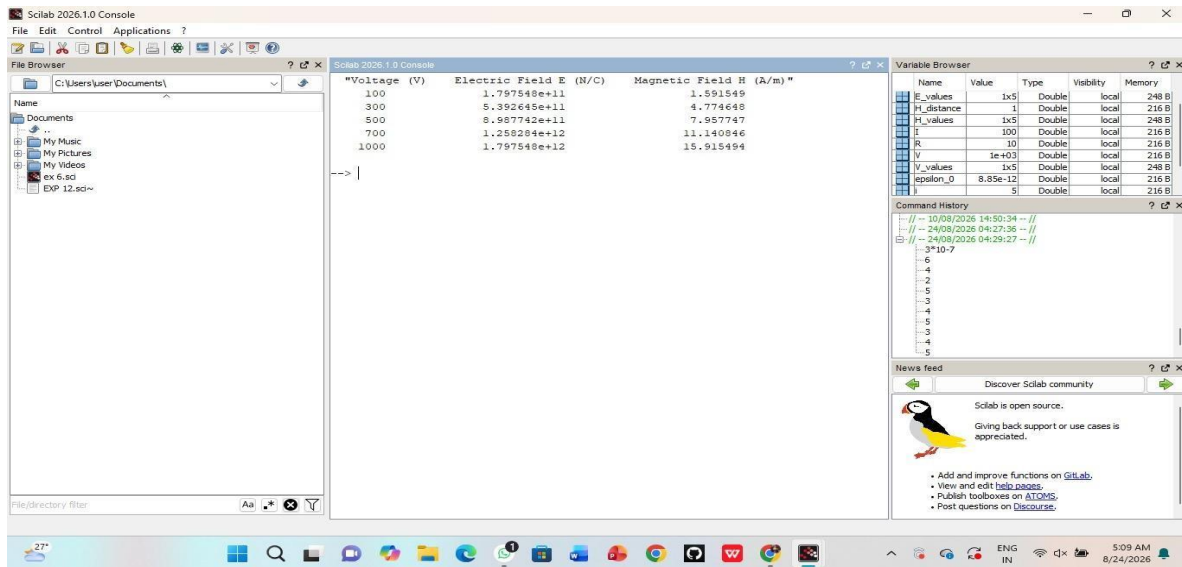


Case 3: To analyze the magnetic and electric field intensities due to an infinitely long conductor when there is a potential difference (voltage) across the conductor, simulating a capacitive effect.

Program:



Output:



Result: Thus the program was verified for the magnetic and electric field intensities due to infinitely long conductor.